

SHIVANSH GOEL

Backend Developer

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PROFESSIONAL SUMMARY

Backend Developer building secure, scalable server-side systems with Node.js, FastAPI, and Python on serverless and edge infrastructure. Experienced designing REST APIs, database schemas, and caching layers, and hardening services with authentication, API-key proxying, and request gating. Skilled at integrating LLM backends and optimizing latency and cost through Redis caching and efficient data access.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript (ES6+), SQL

Backend and Frameworks: Node.js, FastAPI, REST APIs, JSON, CRUD Operations, Server-Side Logic, Cloudflare Workers (Edge Functions)

Databases and Caching: Supabase (PostgreSQL), Cloudflare D1, Redis, MySQL, Query Optimization

API Security and Infrastructure: Access-Key Authentication, API-Key Proxying, LLM Request Gating, Server-Side Model Routing, Rate Control

AI and LLM Backends: LLM Integration, Google Gemini, RAG Pipelines, Embeddings, Structured Output, Prompt Engineering

Tools and Concepts: Git, GitHub, Firebase Tools, Debugging, OOP, DBMS, Computer Networks

PROFESSIONAL EXPERIENCE

Full Stack Developer, Intern

December 2023 – May 2025

Tedforge Solutions Private Limited

India

- Across a 17-month internship, developed Node.js REST APIs powering client-facing web applications and connecting React and Angular frontends to data services.
- Improved MySQL and MongoDB data paths and queries, reducing query overhead and improving response times across product workflows.
- Improved application stability and reliability by reducing reported bugs 40 percent through systematic debugging and refactoring, using Git and GitHub for version control.

PROJECTS

Quizify – AI-Powered Assessment Backend [Live] | *FastAPI, Node.js, Supabase, Redis, Google Gemini, RAG, REST APIs*

- Designed and built the server-side backend for an AI assessment platform, exposing REST endpoints that generate topic-based quizzes on demand from user-provided documentation.
- Implemented a RAG pipeline that chunks, embeds, and retrieves source documents to ground Gemini's generation, with prompts engineered for structured, schema-consistent output.
- Built an AI Security Middleware that proxies all Gemini calls server-side with no client-side API key, authenticating via scoped access keys, gating LLM traffic, and caching results in Redis to cut latency and cost.

GhostRelay – Privacy-First Email Routing Service [Live] [Source] | *Cloudflare Workers, Cloudflare D1, Resend, REST APIs*

- Built the backend of a privacy-first email aliasing service on Cloudflare's edge network, handling alias creation, storage in D1, and inbound email forwarding through Resend.
- Implemented per-alias enable and disable controls and routing logic that forwards mail to a user's real inbox while keeping the real address hidden from spam, breaches, and tracking.

FiTrack AI – Adaptive Fitness Backend [Live] | *Python, Machine Learning, Google Gemini, Node.js, NLP, REST APIs*

- Built the data and AI backend for a fitness platform that analyzes user performance patterns and adaptively adjusts workout intensity.
- Reused a secure server-side proxy pattern to keep Gemini API keys server-side and applied NLP to transform raw performance data into structured, actionable feedback.

Ziplink – URL Shortener SaaS [Live] [Source] | *Next.js, Firebase, Edge Redirects, REST APIs*

- Built the backend of a URL shortener SaaS with edge redirects for low-latency global redirection and Firebase-backed link storage.
- Implemented link creation, QR generation, and per-link click analytics exposed through the application's API.

EDUCATION

Bachelor of Technology in Computer Science and Engineering

2023 – 2027

COER University

Roorkee, India

- Relevant Coursework: Data Structures and Algorithms, Database Management Systems, Operating Systems, Computer Networks, Object-Oriented Programming, Software Engineering.

CERTIFICATIONS

Python 3.11 Programming with FastAPI Backend Framework | **TypeScript 5.2** – Microsoft

Angular 16 with TypeScript 5.2 and PrimeNG 16 | **Onsite Technical Internship – Trainee Software**

Developer (Full Stack)